

INDIA.RUNS

Build what next India runs on

Team Name : White

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Problem Statement : Recruiters go through hundreds of profiles and still often miss the right person. Not because of the talent isn't there – but because keyword filters can't see what actually matters. Build an AI system that ranks candidates the way a great recruiter would – not by matching keywords, but by actually understanding who fits the role

Solution Overview

Proposed Solution

Built a **multi-stage AI candidate ranking engine** that understands both the Job Description and candidate profiles semantically instead of relying only on keyword overlap.

The pipeline combines:

- Keyword Retrieval (BM25)
- Dense Semantic Retrieval (BGE Embeddings)
- Cross-Encoder Re-ranking
- Structured Candidate Intelligence
- Availability Signals
- Explainable Branch reasoning

Why it's different

In Traditional ATS:

- Keyword search only
- Boolean filtering
- Equal importance to every keyword
- No understanding of career quality

Proposed system:

- Understands semantic similarity
- Detects production ML experience
- Evaluates career trajectory
- Detects suspicious profiles
- Considers recruiter interaction signals

JD Understanding & Candidate Evaluation

Intelligent JD Understanding

Instead of embedding the entire JD (which contains company culture, benefits, etc.), we manually convert it into a **focused semantic query** containing only hiring requirements.

Extracted information includes:

- Technical Skills (BM25, Search Systems, Ranking)
- Role Expectations (Production ML systems, System design, Evaluation metrics)
- Business Requirements (Product company background, India preferred, 5–9 years experience)
- Disqualifiers (Entire IT Services career, Frequent job hopping, Outdated ML experience)

Candidate Signals Used:

- Semantic Signals (Skills, Career summaries)
- Structured Signals (Experience fit, ML production experience, Career growth)
- Availability Signals (Notice Period, Response Time, Last Active)
- Quality Signals (Honeypot detection, Career consistency, Verified skill assessments)

Ranking Methodology

Multi-Stage Ranking Pipeline

Stage 1: BM25 retrieval

Retrieve candidates using exact keyword matching.

Stage 2: Dense Retrieval

Uses **BAAI/small-en-v1.5** to generate **384-dimensional** semantic embeddings.

Stage 3: Fusion (Top 2000 candidates)

BM25 and Dense Retrieval each have strengths. Instead of trusting only one, both rankings are fused. This balances lexical and semantic search, is robust to missing keywords, improves recall. We shortlist **top 2000** candidates here.

Stage 4: Structural Scoring (Top 500 candidates)

Each shortlisted candidate receives scores for experience, verified skills, location, company background, carrier trajectory, etc. **Top 500** candidates are shortlisted here.

Stage 5: Cross Encoder Re-ranking (Top 100 candidates)

Uses **cross-encoder/ms-marco-MiniLM-L-6-v2** to jointly compare the JD and candidate for accurate ranking.

Explainability & Data Validation

Explainable Reasoning

Every ranked candidate includes why they matched, which requirements were satisfied, strongest strengths, potential concerns

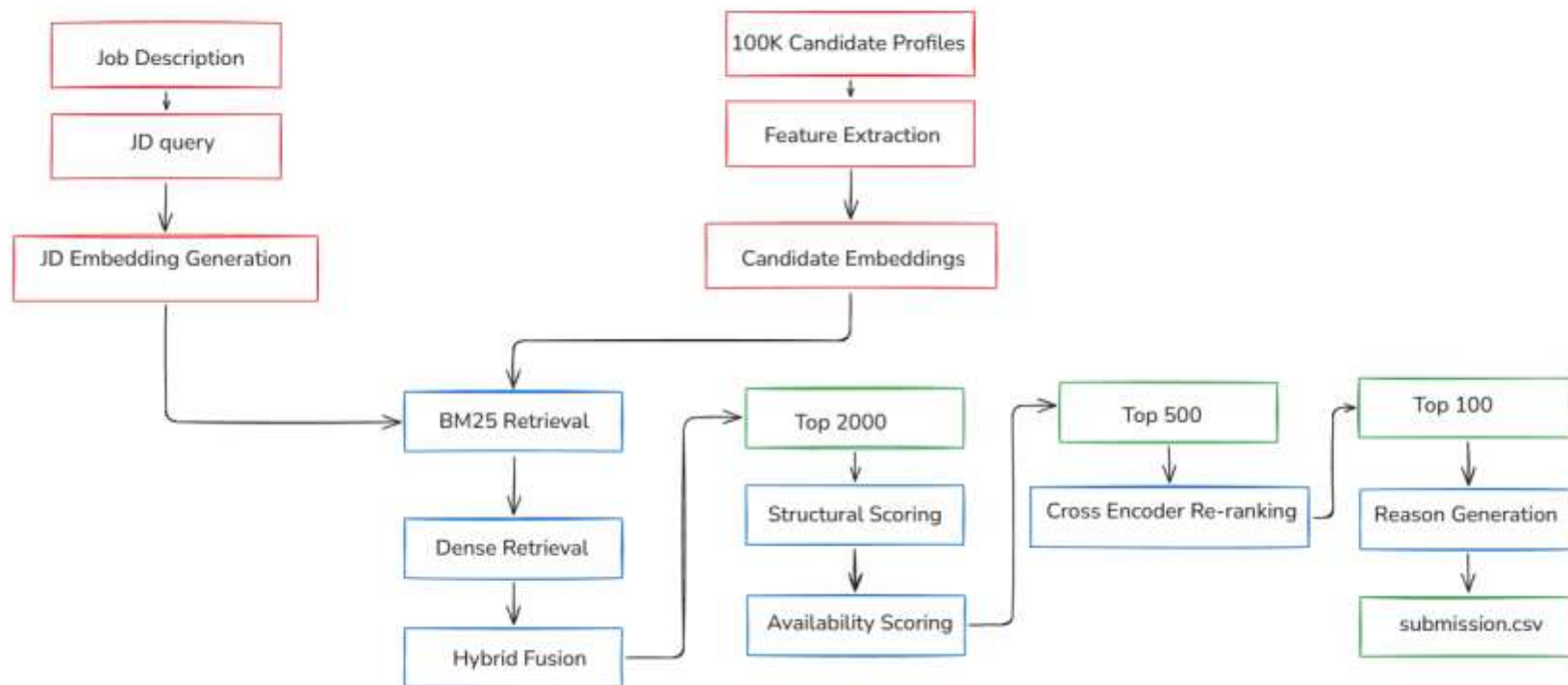
Preventing Hallucinations

Since the reasonings are generated based on multi branch if else statements of reasoning with dynamic insertion of true candidate facts and the reasonings are received based on rankings – Hallucinations never occur since LLMs aren't used

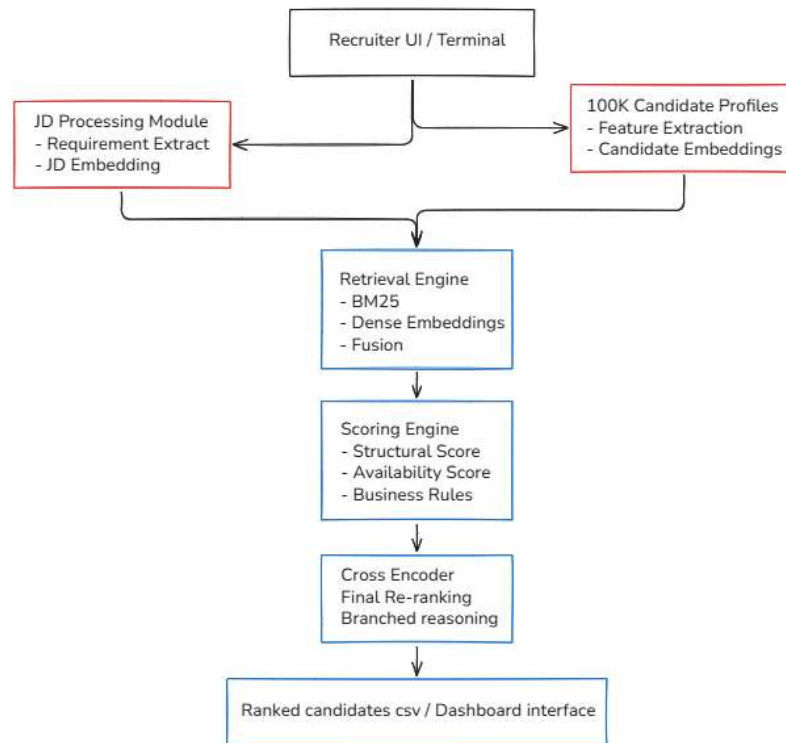
Data Validation

The system detects impossible dates, expert skills with zero experience, career duration inconsistencies , synthetic perfect profiles, entire IT-services careers, suspicious skill inflation - such candidates are heavily penalized or removed before final ranking

End-to-End Workflow



System Architecture



Results & Performance

Runtime

Designed specifically for challenge constraints:

- Ranking takes less than 5 minutes (270s on avg on 8gb ram),

- Needs less than 16gb ram

- Computes everything on CPU only

- Everything works offline, no LLM or API calls needed

Efficiency Strategy

Heavy computation like embeddings, feature extraction, reason generation happens once. Ranking only loads precomputed artifacts making inference extremely fast.

Ranking Quality

Improved quality comes from combining exact keyword search, semantic retrieval, structured candidate intelligence, availability prediction, cross encoder precision

Technologies Used

Technology	Purpose
Python	Pipeline development
BAAI/bge-small-en-v1.5	Dense Retrieval, Embeddings
BM25	Keyword Retrieval
Cross Encoder MiniLM	Final Re-ranking
NumPy	Feature Processing
PyArrow	Parquet Storage
Streamlit	Demo Application
Plotly	Score Visualization
Git LFS	Large Model Artifacts

Submission Assets

Redrob Candidate Ranker — Demo

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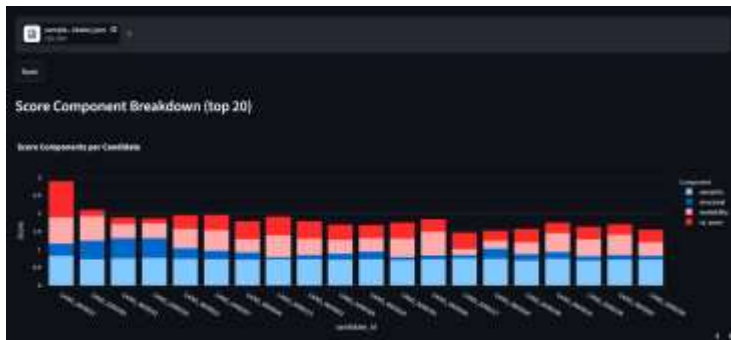
For further information, please contact:

James A. Hughes — founding partner for the production, marketing, sales, and communication agency of Nichols, Davis, & Miller based in New England, has to his credit several national personality magazines, a best-selling weekly, and a communication system for the use of thousands of salesmen — not to mention about 20 years of experience as a member of a major national advertising team. Currently he is writing production code for a new network for adult production and for a new network for children.

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Our Top-10 vs. Keyword-Only Top-10

Full Ranked Output

Proj	Department	Name	Role	Start Date	End Date	Status	Priority	Progress (%)	Notes
1	Engineering	John Doe	Software Engineer	2023-01-15	2023-03-31	Completed	High	100	Project completed successfully.
2	Marketing	Jane Smith	Marketing Specialist	2023-02-01	2023-04-15	In Progress	Medium	75	Campaign launch next week.
3	Finance	Michael Chen	Financial Analyst	2023-03-01	2023-05-31	On Hold	Low	20	Awaiting budget approval.
4	Operations	Sarah Johnson	Operations Manager	2023-01-01	2023-06-30	Active	High	50	Logistics planning in progress.
5	Development	David Kim	Front-end Developer	2023-04-01	2023-07-15	Active	Medium	30	UI design phase complete.
6	QA	Emily White	Quality Assurance	2023-05-01	2023-08-31	Active	Medium	10	Test cases under review.
7	Support	Robert Brown	Customer Support	2023-02-15	2023-09-30	Active	Low	60	Training module development.
8	HR	Lisa Green	Human Resources	2023-03-15	2023-10-31	Active	Medium	40	Recruitment drive for new roles.
9	Legal	James Black	Legal Counsel	2023-04-15	2023-11-30	Active	High	15	Contract review for new vendor.
10	IT	Alice Davis	IT Support	2023-01-01	2023-12-31	Active	Low	80	Network upgrade scheduled.

Reasoning — Top 10

- **IC-0000_000001** - Recommended Safety Engineer (June 0, 2002)
- **IC-0000_000002** - National Engineer (June 0, 2002)
- **IC-0000_000003** - HR Manager (June 0, 2002)
- **IC-0000_000004** - Lead Engineer (June 0, 2002)
- **IC-0000_000005** - National Engineer (June 0, 2002)
- **IC-0000_000006** - Cost Engineer (June 0, 2002)
- **IC-0000_000007** - Technical Engineer (June 0, 2002)
- **IC-0000_000008** - Cost Engineer (June 0, 2002)
- **IC-0000_000009** - Quality Manager (June 0, 2002)
- **IC-0000_000010** - Supply Engineer (June 0, 2002)

Workshop 1.10

GitHub repository: <https://github.com/aayushman-nvm/recruiter-ai>



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THANK YOU